

Solutions of Problems on ch 25

1. Oppositely charged parallel plates are separated by 5.33 mm. A potential difference of 600 V exists between the plates. (a) What is the magnitude of the electric field between the plates? (b) What is the magnitude of the force on an electron between the plates? (c) How much work must be done on the electron to move it to the negative plate if it is initially positioned 2.90 mm from the positive plate?

(a) From Equation 25.6,

$$E = \frac{|\Delta V|}{d} = \frac{600 \text{ J/C}}{5.33 \times 10^{-3} \text{ m}} = \boxed{1.13 \times 10^5 \text{ N/C}}$$

(b) The force on an electron is given by

$$F = |q|E = (1.60 \times 10^{-19} \text{ C})(1.13 \times 10^5 \text{ N/C}) = \boxed{1.80 \times 10^{-14} \text{ N}}$$

(c) Because the electron is repelled by the negative plate, the force used to move the electron must be applied in the direction of the electron's displacement. The work done to move the electron is

$$\begin{aligned} W &= F \cdot s \cos \theta = (1.80 \times 10^{-14} \text{ N})[(5.33 - 2.00) \times 10^{-3} \text{ m}] \cos 0^\circ \\ &= \boxed{4.37 \times 10^{-17} \text{ J}} \end{aligned}$$

2. A uniform electric field of magnitude 250 V/m is directed in the positive x direction. A $+12.0\text{-}\mu\text{C}$ charge moves from the origin to the point $(x, y) = (20.0 \text{ cm}, 50.0 \text{ cm})$. (a) What is the change in the potential energy of the charge–field system? (b) Through what potential difference does the charge move?

(a) We follow the path from (0, 0) to (20.0 cm, 0) to (20.0 cm, 50.0 cm).

$$\Delta U = -(\text{work done})$$

$$\begin{aligned}\Delta U = & -[\text{work from origin to (20.0 cm, 0)}] \\ & - [\text{work from (20.0 cm, 0) to (20.0 cm, 50.0 cm)}]\end{aligned}$$

Note that the last term is equal to 0 because the force is perpendicular to the displacement.

$$\begin{aligned}\Delta U &= -(qE_x)\Delta x = -(12.0 \times 10^{-6} \text{ C})(250 \text{ V/m})(0.200 \text{ m}) \\ &= \boxed{-6.00 \times 10^{-4} \text{ J}}\end{aligned}$$

$$(b) \quad \Delta V = \frac{\Delta U}{q} = -\frac{6.00 \times 10^{-4} \text{ J}}{12.0 \times 10^{-6} \text{ C}} = -50.0 \text{ J/C} = \boxed{-50.0 \text{ V}}$$

5. A uniform electric field $\vec{E}$ of magnitude 325 V/m is directed in the negative y direction in Figure P25.5. The coordinates of point $\textcircled{\text{A}}$ are $(-0.200, -0.300) \text{ m}$, and those of point $\textcircled{\text{B}}$ are $(0.400, 0.500) \text{ m}$. Calculate the electric potential difference $V_{\textcircled{\text{B}}} - V_{\textcircled{\text{A}}}$ using the dashed-line path.

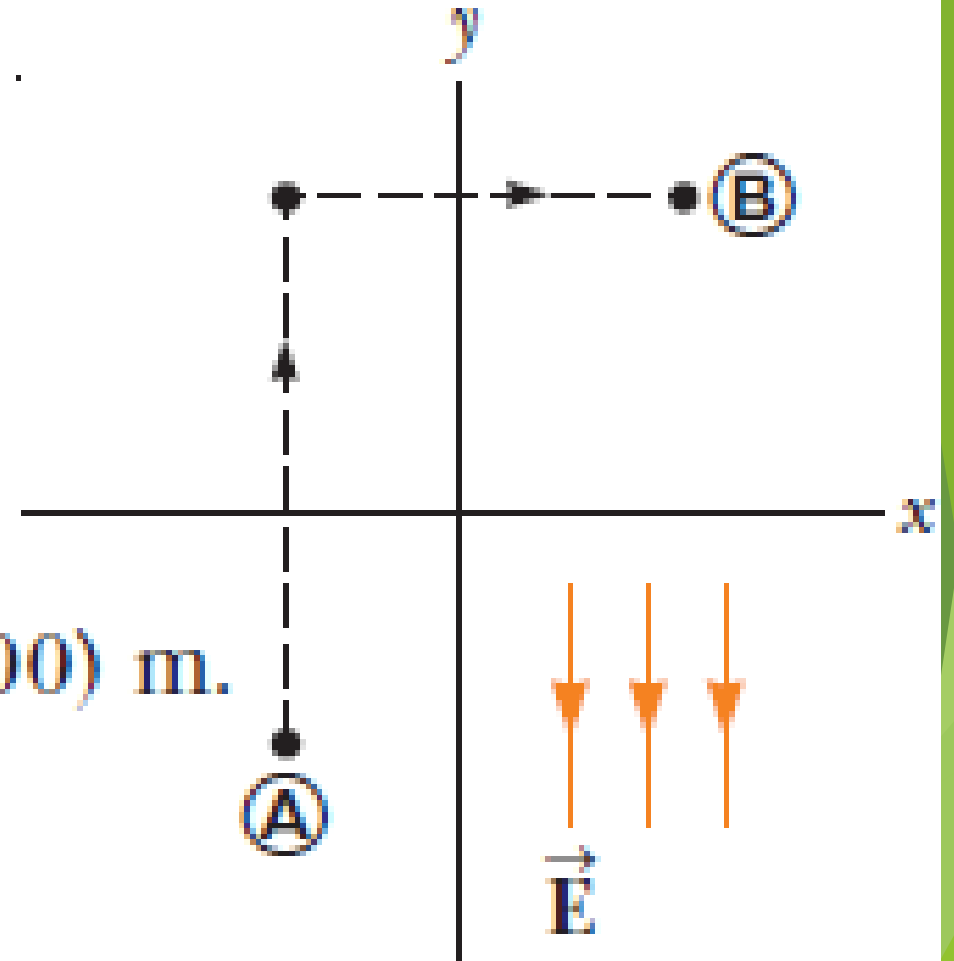


Figure P25.5

$$V_B - V_A = -\int_A^B \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} = -\int_A^C \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} - \int_C^B \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}}$$

$$V_B - V_A = (-E \cos 180^\circ) \int_{-0.300}^{0.500} dy - (E \cos 90.0^\circ) \int_{-0.200}^{0.400} dx$$

$$V_B - V_A = (325 \text{ V/m})(0.800 \text{ m}) = \boxed{+260 \text{ V}}$$

11. An insulating rod having linear charge density $\lambda = 40.0 \mu\text{C/m}$ and linear mass density $\mu = 0.100 \text{ kg/m}$ is released from rest in a uniform electric field $E = 100 \text{ V/m}$ directed perpendicular to the rod (Fig. P25.11). (a) Determine the speed of the rod after it has traveled 2.00 m. (b) What If? How does your answer to part (a) change if the electric field is not perpendicular to the rod? Explain.

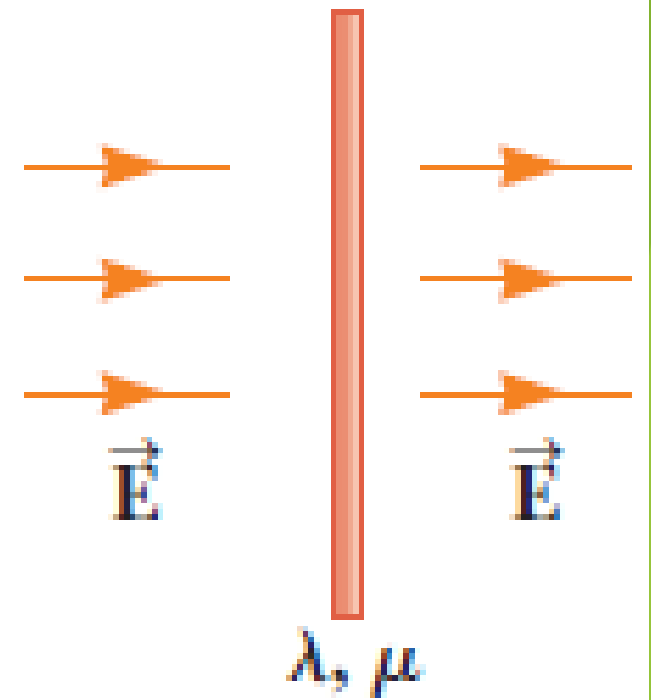


Figure P25.11

Arbitrarily take $V = 0$ at the initial point. Then at distance d downfield, where L is the rod length, $V = -Ed$ and $U_e = -\lambda LEd$.

(a) The rod-field system is isolated:

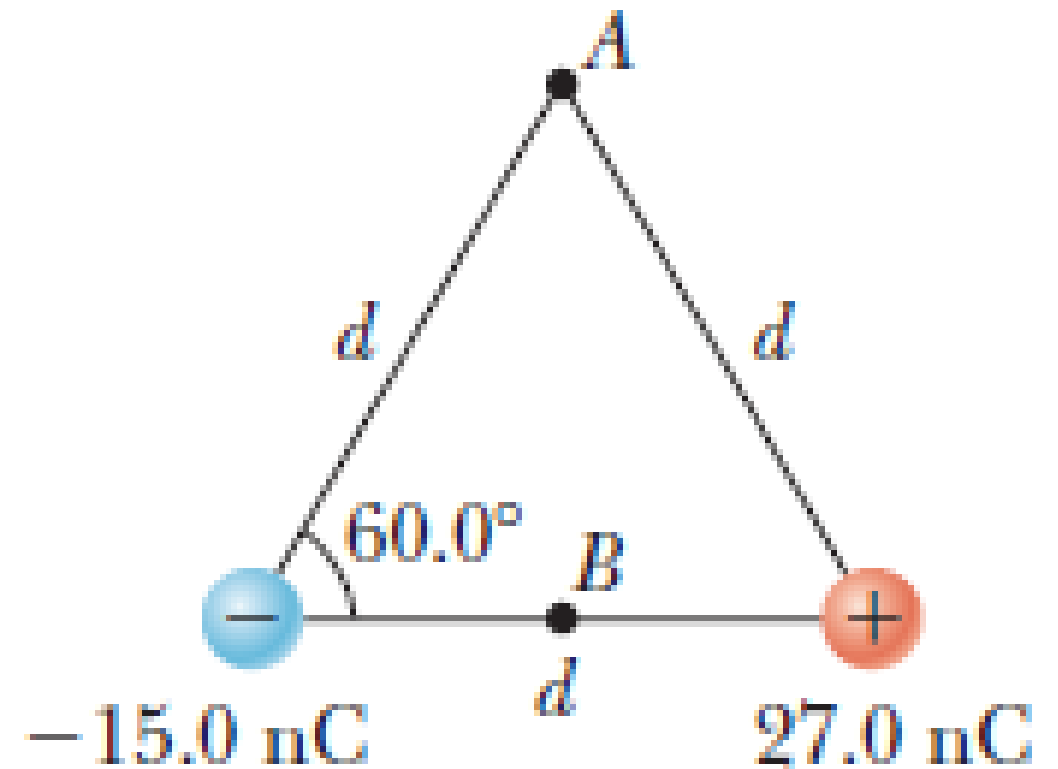
$$K_i + U_i = K_f + U_f \Rightarrow 0 + 0 = \frac{1}{2} m_{\text{rod}} v^2 - qV$$

$$0 = \frac{1}{2} \mu L v^2 - \lambda L E d \Rightarrow \frac{1}{2} \mu L v^2 = \lambda L E d$$

$$v = \sqrt{\frac{2\lambda E d}{\mu}} = \sqrt{\frac{2(40.0 \times 10^{-6} \text{ C/m})(100 \text{ N/C})(2.00 \text{ m})}{(0.100 \text{ kg/m})}}$$
$$= \boxed{0.400 \text{ m/s}}$$

(b) The same. Each bit of the rod feels a force of the same size as before.

14. The two charges in Figure P25.14 are separated by $d = 2.00$ cm. Find the electric potential at (a) point A and (b) point B , which is halfway between the charges.



(a) The electric potential at point A is

$$\begin{aligned} V &= k_e \sum_i \frac{q_i}{r_i} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \\ &\quad \times \left(\frac{-15.0 \times 10^{-9} \text{ C}}{2.00 \times 10^{-2} \text{ m}} + \frac{27.0 \times 10^{-9} \text{ C}}{2.00 \times 10^{-2} \text{ m}} \right) \\ &= \boxed{5.39 \text{ kV}} \end{aligned}$$

(b) The electric potential at point B is

$$\begin{aligned} V &= k_e \sum_i \frac{q_i}{r_i} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \\ &\quad \times \left(\frac{-15.0 \times 10^{-9} \text{ C}}{1.00 \times 10^{-2} \text{ m}} + \frac{27.0 \times 10^{-9} \text{ C}}{1.00 \times 10^{-2} \text{ m}} \right) \\ &= \boxed{10.8 \text{ kV}} \end{aligned}$$

17. Two particles, with charges of 20.0 nC and -20.0 nC , are placed at the points with coordinates $(0, 4.00 \text{ cm})$ and $(0, -4.00 \text{ cm})$ as shown in Figure P25.17. A particle with charge 10.0 nC is located at the origin.

(a) Find the electric potential energy of the configuration of the three fixed charges.

(b) A fourth particle, with a mass of $2.00 \times 10^{-13} \text{ kg}$ and a charge of 40.0 nC , is released from rest at the point $(3.00 \text{ cm}, 0)$. Find its speed after it has moved freely to a very large distance away.

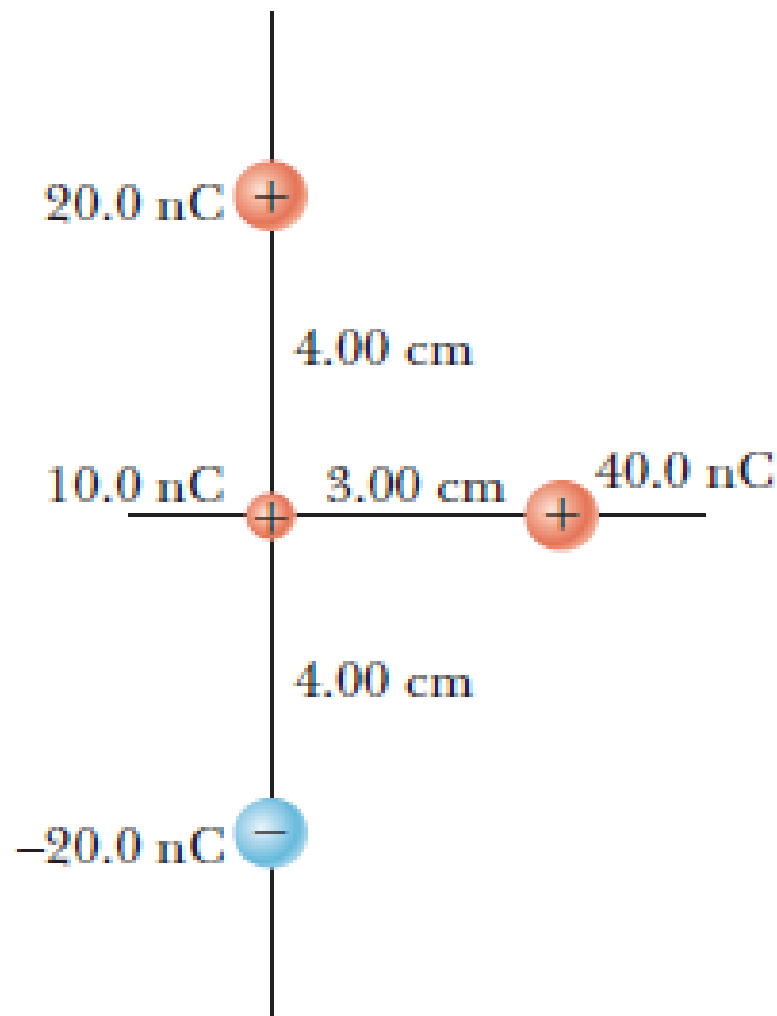


Figure P25.17

(a) In an empty universe, the 20.0-nC charge can be placed at its location with no energy investment. At a distance of 4.00 cm, it creates a potential

$$V_1 = \frac{k_e q_1}{r} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(20.0 \times 10^{-9} \text{ C})}{0.0400 \text{ m}} = 4.50 \text{ kV}$$

To place the 10.0-nC charge there we must put in energy

$$U_{12} = q_2 V_1 = (10.0 \times 10^{-9} \text{ C})(4.50 \times 10^3 \text{ V}) = 4.50 \times 10^{-5} \text{ J}$$

Next, to bring up the -20.0-nC charge requires energy

$$\begin{aligned} U_{23} + U_{13} &= q_3 V_2 + q_3 V_1 = q_3 (V_2 + V_1) \\ &= (-20.0 \times 10^{-9} \text{ C})(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \\ &\quad \times \left(\frac{10.0 \times 10^{-9} \text{ C}}{0.0400 \text{ m}} + \frac{20.0 \times 10^{-9} \text{ C}}{0.0800 \text{ m}} \right) \\ &= -4.50 \times 10^{-5} \text{ J} - 4.50 \times 10^{-5} \text{ J} \end{aligned}$$

The total energy of the three charges is

$$U_{12} + U_{23} + U_{13} = \boxed{-4.50 \times 10^{-5} \text{ J}}$$

(b) The three fixed charges create this potential at the location where the fourth is released:

$$\begin{aligned} V &= V_1 + V_2 + V_3 \\ &= (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \\ &\quad \times \left(\frac{20.0 \times 10^{-9} \text{ C}}{\sqrt{(0.0400 \text{ m})^2 + (0.0300 \text{ m})^2}} \right. \\ &\quad \left. + \frac{10.0 \times 10^{-9} \text{ C}}{0.0300 \text{ m}} - \frac{20.0 \times 10^{-9} \text{ C}}{\sqrt{(0.0400 \text{ m})^2 + (0.0300 \text{ m})^2}} \right) \\ V &= 3.00 \times 10^3 \text{ V} \end{aligned}$$

Energy of the system of four charged objects is conserved as the fourth charge flies away:

$$\begin{aligned} \left(\frac{1}{2}mv^2 + qV \right)_i &= \left(\frac{1}{2}mv^2 + qV \right)_f \\ 0 + (40.0 \times 10^{-9} \text{ C})(3.00 \times 10^3 \text{ V}) &= \frac{1}{2}(2.00 \times 10^{-13} \text{ kg})v^2 + 0 \\ v &= \sqrt{\frac{2(1.20 \times 10^{-4} \text{ J})}{2 \times 10^{-13} \text{ kg}}} = \boxed{3.46 \times 10^4 \text{ m/s}} \end{aligned}$$

18. The two charges in Figure P25.18 are separated by a distance $d = 2.00$ cm, and $Q = +5.00$ nC. Find (a) the electric potential at A , (b) the electric potential at B , and (c) the electric potential difference between B and A .

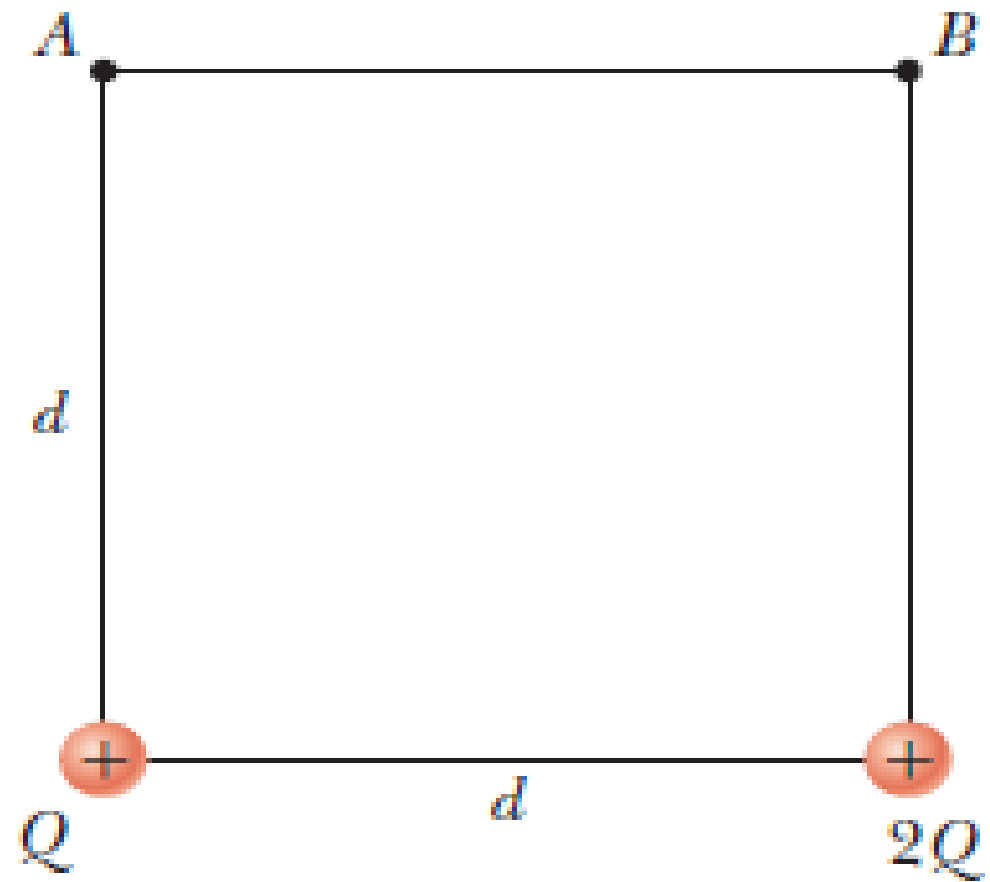


Figure P25.18

$$(a) \quad V_A = k_e \sum_i \frac{q_i}{r_i} = k_e \left(\frac{Q}{d} + \frac{2Q}{d\sqrt{2}} \right) = k_e \frac{Q}{d} (1 + \sqrt{2})$$

$$V = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \left(\frac{5.00 \times 10^{-9} \text{ C}}{2.00 \times 10^{-2} \text{ m}} \right) (1 + \sqrt{2}) = \boxed{5.43 \text{ kV}}$$

$$(b) \quad V_B = k_e \sum_i \frac{q_i}{r_i} = k_e \left(\frac{Q}{d\sqrt{2}} + \frac{2Q}{d} \right) = k_e \frac{Q}{d} \left(\frac{1}{\sqrt{2}} + 2 \right)$$

$$V_B = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \left(\frac{5.00 \times 10^{-9} \text{ C}}{2.00 \times 10^{-2} \text{ m}} \right) \left(\frac{1}{\sqrt{2}} + 2 \right) = \boxed{6.08 \text{ kV}}$$

$$(c) \quad V_B - V_A = k_e \frac{Q}{d} \left(\frac{1}{\sqrt{2}} + 2 \right) - k_e \frac{Q}{d} (1 + \sqrt{2}) = k_e \frac{Q}{d} \left(\frac{1}{\sqrt{2}} + 1 - \sqrt{2} \right)$$

$$V_B - V_A = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \left(\frac{5.00 \times 10^{-9} \text{ C}}{2.00 \times 10^{-2} \text{ m}} \right) \left(\frac{1}{\sqrt{2}} + 1 - \sqrt{2} \right) \\ = \boxed{658 \text{ V}}$$

23. A particle with charge $+q$ is at the origin. A particle with charge $-2q$ is at $x = 2.00$ m on the x axis. (a) For what finite value(s) of x is the electric field zero? (b) For what finite value(s) of x is the electric potential zero?

$$(a) \quad E_x = \frac{k_e q_1}{x^2} + \frac{k_e q_2}{(x - 2.00)^2} = 0 \quad \text{becomes} \quad E_x = k_e \left(\frac{+q}{x^2} + \frac{-2q}{(x - 2.00)^2} \right) = 0$$

$$\text{Dividing by } k_e, \quad 2qx^2 = q(x - 2.00)^2$$

$$\text{or} \quad x^2 + 4.00x - 4.00 = 0.$$

$$\text{Therefore } E = 0 \quad \text{when} \quad x = \frac{-4.00 \pm \sqrt{16.0 + 16.0}}{2} = -\boxed{4.83 \text{ m}}.$$

(Note that the positive root does not correspond to a physically valid situation.)

(b) Assuming $0 < x < 2.00$ m, the potential is zero when

$$V = \frac{k_e q_1}{x} + \frac{k_e q_2}{2.00 - x} = 0 \quad \text{or} \quad \frac{V}{k_e} = \left[\frac{(q)}{x} + \frac{(-2q)}{2.00 - x} \right] = 0$$

giving $q(2.00 - x) = 2qx$ or $x = \frac{2.00}{3} = \boxed{0.667 \text{ m}}$

For $x > 2.00 \text{ m}$, the potential is zero when

$$\frac{V}{k_e} = \left[\frac{(q)}{x} + \frac{(-2q)}{x - 2.00} \right] = 0 \quad \text{or} \quad q(x - 2.00) = 2qx$$

This has no positive x solution. Physically, the total potential cannot be zero for any point where $x > 2.00 \text{ m}$ because that point is closer to charge $-2q$, so its potential is always more negative than the potential from charge q is positive. For $x < 0$, the

potential is zero when $\frac{V}{k_e} = \left[\frac{(q)}{|x|} + \frac{(-2q)}{|2.00 + x|} \right] = 0$, giving

$$\frac{q}{|x|} < \frac{2q}{2.00 + |x|} \quad \text{or} \quad q(2.00 + |x|) = 2q|x|$$

which has the solution $|x| = 2.00$ correspond to $x = \boxed{-2.00 \text{ m}}$.

27. Four identical charged particles ($q = +10.0 \mu\text{C}$) are **W** located on the corners of a rectangle as shown in Figure P25.27. The dimensions of the rectangle are $L = 60.0 \text{ cm}$ and $W = 15.0 \text{ cm}$. Calculate the change in electric potential energy of the system as the particle at the lower left corner in Figure P25.27 is brought to this position from infinitely far away. Assume the other three particles in Figure P25.27 remain fixed in position.

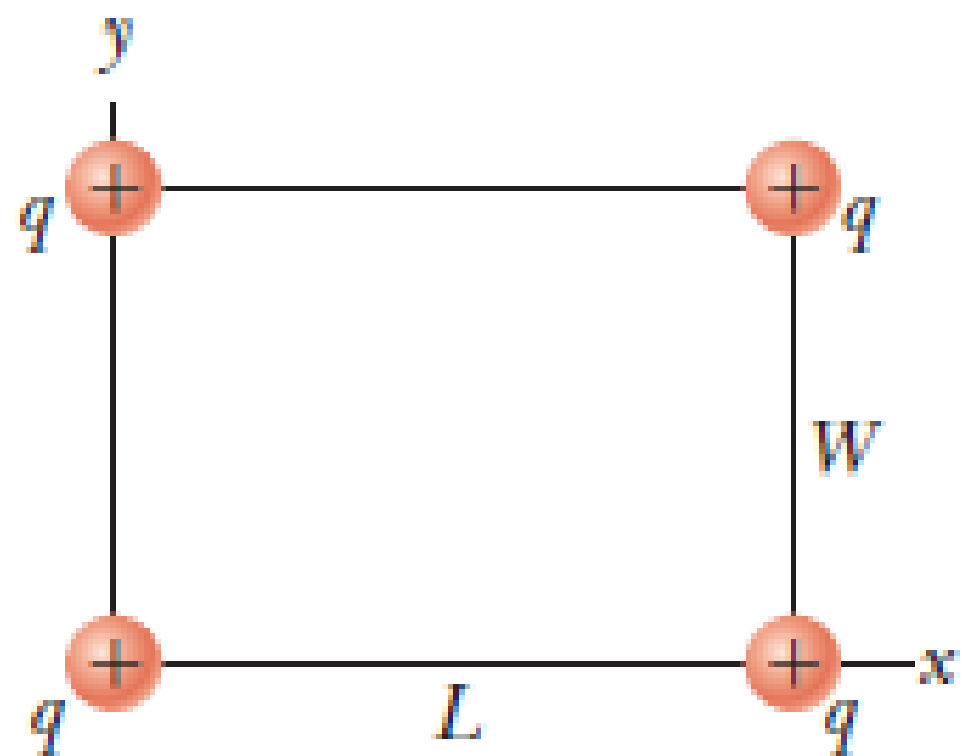


Figure P25.27

The total change in potential energy is the sum of the change in potential energy of the $q_1 - q_4$, $q_2 - q_4$, and $q_3 - q_4$ particle systems:

$$U_e = q_4 V_1 + q_4 V_2 + q_4 V_3 = q_4 k_e \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \right)$$

$$U_e = (10.0 \times 10^{-6} \text{ C})^2 (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)$$

$$\times \left(\frac{1}{0.600 \text{ m}} + \frac{1}{0.150 \text{ m}} + \frac{1}{\sqrt{(0.600 \text{ m})^2 + (0.150 \text{ m})^2}} \right)$$

$$U_e = \boxed{8.95 \text{ J}}$$

29. Five particles with equal negative charges $-q$ are placed symmetrically around a circle of radius R . Calculate the electric potential at the center of the circle.

Each charge creates equal potential at the center. The total potential is

$$V = 5 \left[\frac{k_e (-q)}{R} \right] = \boxed{-\frac{5k_e q}{R}}$$

33. How much work is required to assemble eight identical charged particles, each of magnitude q , at the corners of a cube of side s ?

A cube has 12 edges and 6 faces. Consequently, there are 12 edge pairs separated by s , $2 \times 6 = 12$ face diagonal pairs separated by $\sqrt{2}s$, and 4 interior diagonal pairs separated by $\sqrt{3}s$.

$$U = \frac{k_e q^2}{s} \left[12 + \frac{12}{\sqrt{2}} + \frac{4}{\sqrt{3}} \right] = \boxed{22.8 \frac{k_e q^2}{s}}$$

36. Figure P25.36 represents a graph of the electric potential in a region of space versus position x , where the electric field is parallel to the x axis. Draw a graph of the x component of the electric field versus x in this region.

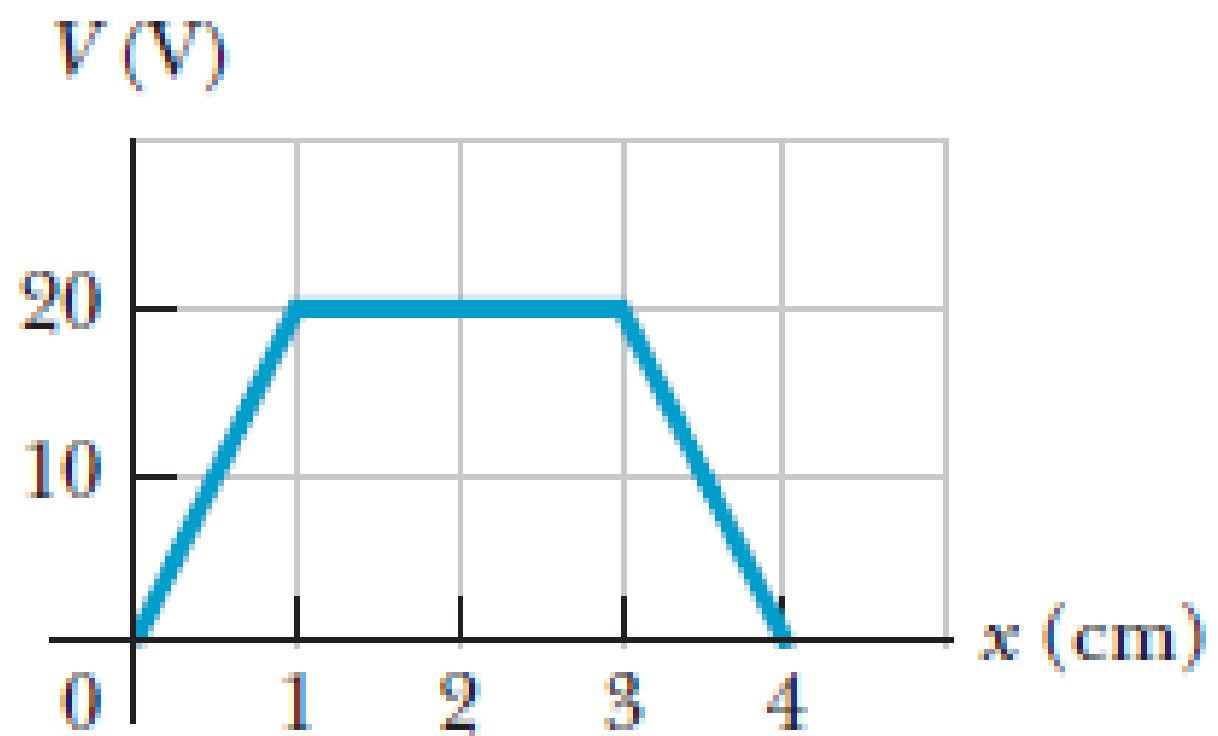


Figure P25.36

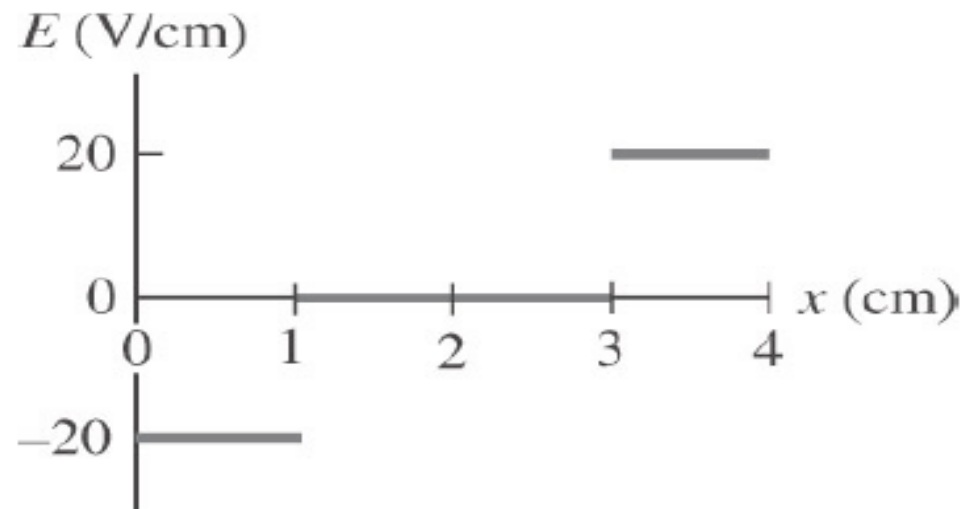
$$E_x = -\frac{\partial V}{\partial x} \rightarrow E_x = -\frac{\Delta V}{\Delta x} = -(\text{slope of line})$$

The sign indicates the direction of the x component of the field.

$$x = 0 \text{ to } 1 \text{ cm:} \quad E_x = -\frac{\Delta V}{\Delta x} = -\frac{20 \text{ V} - 0}{1 \text{ cm}} = -20 \text{ V/cm}$$

$$x = 1 \text{ to } 3 \text{ cm:} \quad E_x = -\frac{\Delta V}{\Delta x} = -\frac{0}{2 \text{ cm}} = 0 \text{ V/m}$$

$$x = 3 \text{ to } 4 \text{ cm:} \quad E_x = -\frac{\Delta V}{\Delta x} = -\frac{0 - 20 \text{ V}}{1 \text{ cm}} = +20 \text{ V/cm}$$



37. The potential in a region between $x = 0$ and $x = 6.00$ m is $V = a + bx$, where $a = 10.0$ V and $b = -7.00$ V/m. Determine (a) the potential at $x = 0$, 3.00 m, and 6.00 m and (b) the magnitude and direction of the electric field at $x = 0$, 3.00 m, and 6.00 m.

$$V = a + bx = 10.0 \text{ V} + (-7.00 \text{ V/m})x$$

(a) At $x = 0$, $V = \boxed{10.0 \text{ V}}$

At $x = 3.00 \text{ m}$, $V = \boxed{-11.0 \text{ V}}$

At $x = 6.00 \text{ m}$, $V = \boxed{-32.0 \text{ V}}$

(b) $E = -\frac{dV}{dx} = -b = -(-7.00 \text{ V/m}) = \boxed{7.00 \text{ N/C in the } +x \text{ direction}}$

39. Over a certain region of space, the electric potential is $V = 5x - 3x^2y + 2yz^2$. (a) Find the expressions for the x , y , and z components of the electric field over this region. (b) What is the magnitude of the field at the point P that has coordinates $(1.00, 0, -2.00)$ m?

(a) $V = 5x - 3x^2y + 2yz^2$, where x , y and z are in meters and V is in volts.

$$E_x = -\frac{\partial V}{\partial x} = -5 + 6xy$$

$$E_y = -\frac{\partial V}{\partial y} = +3x^2 - 2z^2$$

$$E_z = -\frac{\partial V}{\partial z} = -4yz$$

which gives

$$\vec{E} = (-5 + 6xy)\hat{i} + (3x^2 - 2z^2)\hat{j} - 4yz\hat{k}$$

(b) Evaluate E at $(1.00, 0, -2.00)$ m, suppressing units,

$$E_x = -5 + 6(1.00)(0) = -5.00$$

$$E_y = 3(1.00)^2 - 2(-2.00)^2 = -5.00$$

$$E_z = -4(0)(-2.00) = 0$$

which gives

$$E = \sqrt{E_x^2 + E_y^2 + E_z^2} = \sqrt{(-5.00)^2 + (-5.00)^2 + 0^2} = \boxed{7.07 \text{ N/C}}$$

49. How many electrons should be removed from an initially uncharged spherical conductor of radius 0.300 m to produce a potential of 7.50 kV at the surface?

Substituting given values into $V = \frac{k_e Q}{r}$, with $Q = Nq$:

$$7.50 \times 10^3 \text{ V} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)Q}{0.300 \text{ m}}$$

Substituting $q = 2.50 \times 10^{-7} \text{ C}$,

$$N = \frac{2.50 \times 10^{-7} \text{ C}}{1.60 \times 10^{-19} \text{ C}/e^-} = \boxed{1.56 \times 10^{12} \text{ electrons}}$$

50. A spherical conductor has a radius of 14.0 cm and a charge of $26.0 \mu\text{C}$. Calculate the electric field and the electric potential at (a) $r = 10.0 \text{ cm}$, (b) $r = 20.0 \text{ cm}$, and (c) $r = 14.0 \text{ cm}$ from the center.

M

- (a) Inside a conductor when charges are not moving, the electric field is zero and the potential is uniform, the same as on the surface, and $E = \boxed{0}$.

$$V = \frac{k_e q}{R} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(26.0 \times 10^{-6} \text{ C})}{0.140 \text{ m}} = \boxed{1.67 \text{ MV}}$$

(b)
$$E = \frac{k_e q}{r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(26.0 \times 10^{-6} \text{ C})}{(0.200 \text{ m})^2}$$
$$= \boxed{5.84 \text{ MN/C}} \text{ away}$$

$$V = \frac{k_e q}{R} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(26.0 \times 10^{-6} \text{ C})}{0.200 \text{ m}} = \boxed{1.17 \text{ MV}}$$

(c)
$$E = \frac{k_e q}{R^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(26.0 \times 10^{-6} \text{ C})}{(0.140 \text{ m})^2}$$
$$= \boxed{11.9 \text{ MN/C}} \text{ away}$$

$$V = \frac{k_e q}{R} = \boxed{1.67 \text{ MV}}$$